Gated Engineering Plan for Million-Qubit Modular Quantum Computing: A Scenario-Based Roadmap Through 2030

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Abstract—This capstone paper presents a gated engineering plan for the QONTOS modular quantum computing program through 2030. Rather than a single-path narrative toward one million physical qubits, we define three explicit scenarios—Base, Aggressive, and Stretch—each with measurable validation gates, quantitative no-go conditions, and concrete fallback plans. The canonical architecture proceeds through four integration levels: chiplet (approximately 2,000 qubits), module (approximately 10,000 qubits), system (approximately 100,000 qubits), and data center (approximately 1,000,000 qubits). Each level is gated by subsystem milestones in device performance, quantum error correction (QEC) overhead, photonic interconnects, cryogenic infrastructure, and classical control. We synthesize requirements from the nine preceding papers in the QONTOS series and map them onto a five-phase program with explicit decision points. Under aggressive but explicit assumptions regarding device coherence, QEC overhead reduction, and interconnect fidelity, QONTOS could reach quantum advantage on selected applications within the 2030 timeframe. The stretch scenario of 10,000 logical qubits remains a conditional north star, reachable only if every major subsystem achieves its aggressive-case milestone on schedule.

[CLAIM-ROADMAP] This document is a scenario-based engineering plan. All targets are conditional on stated subsystem milestones and are not unconditional predictions.

I. Introduction and Roadmap Philosophy

A. Motivation

The quantum computing industry has entered a phase where credible roadmaps must move beyond aspirational qubit counts toward engineering plans with explicit decision logic. IBM has published development roadmaps projecting 100,000-qubit systems through modular architectures [1]. Google Quantum AI has demonstrated error correction below the surface code threshold [2]. These milestones confirm that large-scale fault-tolerant quantum computing is an engineering challenge with identifiable subsystem dependencies, not a speculative ambition.

[CLAIM-CONTEXT] The transition from NISQ-era demonstrations [10] to fault-tolerant systems requires systematic planning with quantitative gates, not aspirational timelines.

However, as the National Academies consensus study emphasized, scaling to practical quantum advantage requires simultaneous advances across multiple subsystems—qubits, error correction, interconnects, control, and software—with failure in any single domain potentially blocking the entire program [8]. This reality demands scenario-based planning.

B. Roadmap Design Principles

This roadmap is built on four principles:

1) Scenario discipline. Every phase specifies Base, Aggressive, and Stretch outcomes.
2) Quantitative gates. Advancement requires meeting measurable subsystem milestones.
3) Explicit no-go conditions. Each phase defines conditions under which advancement is blocked and fallback plans are activated.
4) Cross-subsystem coordination. A qubit milestone without a corresponding QEC or interconnect milestone does not constitute phase completion.

C. Scenario Summary

TABLE I
Scenario summary.

Scenario	2030 Target	Governing Assumption
Base	10,000–20,000 physical qubits; 10–20 logical qubits	Current trajectories with incremental improvement
Aggressive	100,000 physical qubits; 100–1,000 logical qubits	Major but plausible breakthroughs on schedule
Stretch	1,000,000 physical qubits; 10,000 logical qubits	Every subsystem achieves aggressive-case milestone

[CLAIM-PROJECTION] The Stretch scenario is not a baseline expectation. It is a conditional planning target that requires simultaneous success across all subsystem domains.

D. Canonical Architecture Hierarchy

The QONTOS architecture [13] follows a four-level integration hierarchy:

TABLE II
Architecture integration hierarchy.

Level	Name	Qubits	Challenge
L1	Chiplet	∼2,000	On-chip wiring, yield
L2	Module	∼10,000	Multi-chiplet packaging
L3	System	∼100,000	Photonic links, dist. QEC
L4	Data Center	∼1,000,000	Multi-fridge networking

This hierarchy is consistent with modular architectures proposed by Monroe et al. [4] and reflected in IBM's modular quantum roadmap [1].

II. Subsystem Milestone Framework

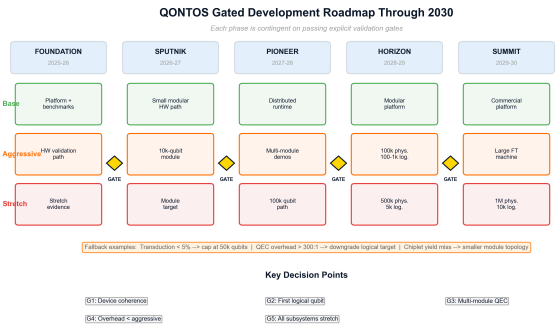

Fig. 1. Gated Development Roadmap.

Before defining program phases, we establish the subsystem milestones that gate phase transitions.

A. Device Performance Milestones (from Paper 2)

TABLE III
Device performance milestones.

ID	Metric	Base	Aggr.	Stretch
D-1	T_1 relaxation	300 µs	800 µs	>1 ms
D-2	T_2 coherence	200 µs	500 µs	>800 µs
D-3	1Q gate error	10^{-3}	5×10^{-4}	10^{-4}
D-4	2Q gate error	5×10^{-3}	10^{-3}	5×10^{-4}
D-5	Chiplet yield	>85%	>92%	>96%

[CLAIM-ENGINEERING] Device milestones D-1 through D-4 are informed by demonstrated tantalum-on-silicon transmon performance. Kim et al. [12] demonstrated utility-scale quantum computation with comparable gate fidelities.

B. QEC Milestones (from Paper 3)

TABLE IV
Quantum error correction milestones.

ID	Metric	Base	Aggr.	Stretch
Q-1	QEC overhead	1,000:1	300:1	100:1
Q-2	Logical error/round	10^{-4}	10^{-6}	10^{-8}
Q-3	Code distance	d=5	d=9	d=13
Q-4	Decoder latency	<10 µs	<1 µs	<500 ns

[CLAIM-ENGINEERING] The Stretch overhead target of 100:1 is informed by Bravyi et al. [3]. The Aggressive 300:1 target reflects surface-code implementations consistent with analyses by Chamberland et al. [11].

C. Interconnect Milestones (from Paper 4)

TABLE V
Interconnect milestones.

ID	Metric	Base	Aggr.	Stretch
I-1	Transduction eff.	<1%	5–15%	>25%
I-2	Bell pair fidelity	85%	95%	>99%
I-3	Bell pair rate	1 kHz	100 kHz	>1 MHz
I-4	Links per module	1–2	4–8	16+

[CLAIM-ENGINEERING] Transduction efficiency is the single highest-risk interconnect parameter. Current laboratory demonstrations remain below 5% end-to-end efficiency.

D. Cryogenic and Control Milestones (from Papers 5 and 7)

TABLE VI
Cryogenic and control milestones.

ID	Metric	Base	Aggr.	Stretch
C-1	Cooling at 20 mK	20 µW	50 µW	>100 µW
C-2	Control lines/module	5,000	15,000	50,000+
C-3	Cryo mux ratio	4:1	16:1	64:1
C-4	Power/phys. qubit	50 W	15 W	5 W

E. Software and Algorithmic Milestones (from Papers 6 and 8)

TABLE VII
Software and algorithmic milestones.

ID	Metric	Base	Aggr.	Stretch
S-1	Compilation latency	<60 s	<5 s	<1 s
S-2	Distributed overhead	>50%	<20%	<5%
S-3	Replay fidelity	90%	98%	>99.5%
S-4	Circuits demonstrated	Variational	FT subroutines	Full FTQC

III. Five-Phase Gated Program

The program proceeds through five named phases, each approximately 12–18 months. Phase transitions require

meeting quantitative validation gates; failure triggers explicit fallback plans.

A. Phase 1: FOUNDATION (2025–2026)

Primary Objective: Establish the canonical hybrid superconducting-photonic architecture, software platform, benchmarking methodology, and device characterization baseline.

TABLE VIII
Phase 1 scenario outcomes.

Dim.	Base	Aggressive	Stretch Rel.
HW	Param. survey done; chiplet frozen	First 2k-qubit chiplet	Yield data supports L1
QEC	Surface code sim. at d=3	d=5 logical qubit	Overhead consistent with Q-1
SW	Orchestration v1.0	HW-in-loop sim. validated	DT predicts within 10%
Bench.	Methodology published	Ladder through L2	Targets set for all levels

Validation Gates:

- GATE-F1: Architecture constants frozen and documented.
- GATE-F2: Benchmark methodology peer-reviewed and reproducible.
- GATE-F3: Device assumptions populated with literature-grounded values.
- GATE-F4: Software platform demonstrates end-to-end compilation and execution.

No-Go Conditions: No coherent benchmark program after 12 months; device literature reveals no path to D-1 Base; software cannot compile for >500 qubits. Fallback: Continue as software-first platform with external hardware partners.

B. Phase 2: SPUTNIK (2026–2027)

Primary Objective: Validate chiplet-to-module integration and demonstrate first logical qubits.

TABLE IX
Phase 2 scenario outcomes.

Dim.	Base	Aggressive	Stretch Rel.
HW	Multi-chiplet packaging; 5k-qubit partial	Full 10k-qubit module, >85% yield	Yield supports L2
QEC	d=5, overhead <2,000:1	d=7, overhead ∼500:1	Trajectory to Q-1 stretch
Intercon.	Intra-module coupling	First inter-module plan	Transduction 3–5%
Cryo	1 fridge, 1 module	Thermal budget for 2 modules	C-1 base met

Validation Gates: GATE-S1 (chiplet yield >85%), GATE-S2 (logical qubit lifetime > 10× physical T_1), GATE-S3 (module thermal budget feasible), GATE-S4 (inter-module experiment design review passed).

No-Go/Fallback: Chiplet yield <80%: pivot to smaller module topology. QEC overhead >3,000:1: invest in device improvement. Thermal budget exceeds capacity by >2×: reduce module population.

C. Phase 3: PIONEER (2027–2028)

Primary Objective: Validate multi-module distributed execution and demonstrate fault-tolerant subroutines.

TABLE X
Phase 3 scenario outcomes.

Dim.	Base	Aggressive	Stretch Rel.
HW	2-module (20k Q)	5-module (50k Q)	Path to L3
QEC	Dist. surface code, 2 mod.	Dist. d=9, ∼300:1	Q-1 stretch trajectory
Intercon.	Bell >1 kHz	Fidelity >93%, >10 kHz	Transd. >10%
SW	2-module runtime	5-module compiler opt.	Overhead <20%
Algo.	Variational on logical Q	First FT subroutine	Flagship validated

Validation Gates: GATE-P1 (Bell pair fidelity >85%, rate >1 kHz), GATE-P2 (distributed surface code within 2× single-module), GATE-P3 (decoder latency <10 µs), GATE-P4 (at least one FT subroutine executed).

[CLAIM-ENGINEERING] The transduction gate is the single highest-risk decision point in the entire roadmap. If efficiency stays below 5%, photonic multi-system networking is not viable within 2030. Fallback: Direct microwave inter-module coupling, limiting scale to ∼50,000 qubits.

D. Phase 4: HORIZON (2028–2029)

Primary Objective: Scale to system-level integration (100,000 qubits) and demonstrate quantum advantage on targeted applications.

TABLE XI
Phase 4 scenario outcomes.

Dim.	Base	Aggressive	Stretch Rel.
HW	3–5 mod. (30–50k Q)	10-mod. (100k Q)	Path to L4
QEC	$\epsilon_L < 10^{-4}$ at d=7	$\epsilon_L < 10^{-6}$ at d=9; 300:1	On track for 100:1
Intercon.	Stable >10 kHz	Fidelity >97%, >100 kHz	Multi-fridge link
Apps.	Small mol. outperforms classical	Phase est. on relevant mol.	Resource est. validated
SW	Production orchestration	Auto partitioning <10% overhead	Full-stack optimization

Validation Gates: GATE-H1 (QEC overhead ≤ 300:1), GATE-H2 (at least one application shows quantum advantage), GATE-H3 (multi-fridge photonic link if pursuing

Stretch), GATE-H4 (monotonic benchmark improvement across three campaigns).

[CLAIM-PROJECTION] Reiher et al. [5] estimated that simulating the FeMoco active site requires approximately 4,000 logical qubits with T-gate counts in the billions. This remains a stretch-scenario flagship, feasible only if both QEC overhead and logical gate rates meet aggressive targets.

E. Phase 5: SUMMIT (2029–2030)

Primary Objective: Determine whether the Stretch architecture is reachable; deliver the strongest achievable system.

TABLE XII
Phase 5 scenario outcomes.

Dim.	Base	Aggressive	Stretch
HW	50k Q, 5+ mod.	100k+ Q, 10+ mod.	1M Q, 100+ mod.
Logical Q	10–50	100–1,000	10,000
QEC	∼1,000:1	∼300:1	∼100:1
Apps.	Useful sims.; hybrid	FT on flagship	Advantage on catalysis
Commercial	Cloud platform	Enterprise service	Quantum data center

Validation Gates: GATE-SM1 (device milestones at Aggressive+), GATE-SM2 (QEC overhead at target), GATE-SM3 (interconnect milestones at Aggressive+), GATE-SM4 (flagship application advantage with audit trail), GATE-SM5 (>168 hours continuous operation).

Stretch-Scenario Requirements: ALL must hold simultaneously: (1) physical qubits >500,000; (2) QEC overhead $\leq$ 150:1; (3) transduction efficiency >20%; (4) at least one application advantage that cannot be replicated classically within 1,000× the time; (5) full software stack at data-center scale.

[CLAIM-ROADMAP] Failure of any single condition triggers fallback to the Aggressive architecture.

IV. Scenario Planning Tables

A. Architecture Scenario Table

TABLE XIII
Architecture scenario comparison (2030).

Parameter	Base	Aggr.	Stretch
Physical qubits	10–20k	100k	1M
Logical qubits	10–20	100–1k	10k
QEC overhead	∼1,000:1	∼300:1	∼100:1
Arch. level	L2	L3	L4
Modules	1–2	10	100+
Link type	MW direct	Photonic intra-fridge	Photonic inter-fridge
Fridges	1	1–2	10+
Decoder	SW MWPM	FPGA-accel.	ASIC / neural
Flagship	Variational	FT subroutines	Catalysis / crypt.

B. Investment Scenario Table

TABLE XIV
Investment planning envelopes.

Category	Base	Aggr.	Stretch
Hardware R&D	\$30–50M	\$100–150M	\$200–300M
Fabrication	\$10–20M	\$50–80M	\$150–200M
Cryogenics	\$5–10M	\$20–40M	\$80–120M
Software	\$15–25M	\$30–50M	\$50–80M
Operations	\$5–10M	\$15–25M	\$50–80M
Total	\$65–115M	\$215–345M	\$530–780M

[CLAIM-ESTIMATE] Investment figures are planning envelopes informed by McKinsey's Quantum Technology Monitor [9] and publicly available data from comparable programs.

C. Application Scenario Table

TABLE XV
Application scenario mapping.

Domain	Base	Aggr.	Stretch
Chemistry	Small-mol. VQE	FT phase est.	Full FeMoco [5]
Materials	Variational lattice	Hubbard at scale	Correlated electrons
Optimization	QAOA toys	Hybrid QC opt.	Grover-accel. search
Cryptography	Shor <20-bit	Shor 256-bit	RSA-2048 [7]
ML	Quantum kernels	QE feature spaces	Provable advantage

[CLAIM-CONTEXT] Gidney and Ekera [7] estimated RSA-2048 factoring requires approximately 20 million noisy qubits, placing this firmly in the Stretch scenario. Beverland et al. [6] confirm that most applications of economic interest require >1,000 logical qubits.

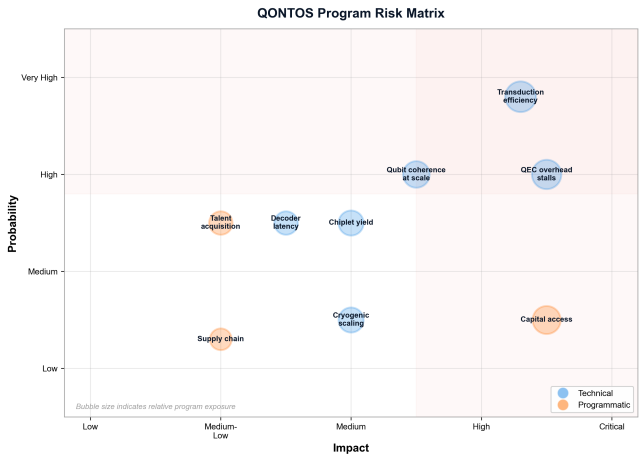

Fig. 2. Program Risk Matrix.

V. Risk Matrix

A. Technical Risk Assessment

TABLE XVI
Technical risk assessment.

ID	Risk	Prob.	Impact	Scenario
R-01	Transduction plateaus <5%	High	Critical	Str., Aggr.
R-02	QEC overhead >500:1	Med.	High	Stretch
R-03	Chiplet yield <80%	Med.	High	All
R-04	Decoder latency too high	Med.	High	Aggr., Str.
R-05	Cryo cooling insufficient	Med.	Med.	Stretch
R-06	Control wiring limits	Med.	Med.	Stretch
R-07	No advantage by 2029	Med.	High	Aggr., Str.
R-08	Supply chain disruption	Low	High	All
R-09	Key personnel departure	Med.	Med.	All
R-10	Competitor achieves first	Low	Med.	All

B. Programmatic Risk Assessment

TABLE XVII
Programmatic risk assessment.

ID	Risk	Prob.	Impact
P-01	Capital access interrupted	Low–Med.	Critical
P-02	Fab partner constraints	Med.	High
P-03	Regulatory changes	Low	Med.
P-04	Timeline optimism bias	High	Med.

VI. Cross-Paper Dependencies

A. Paper-to-Phase Dependency Matrix

TABLE XVIII
Paper-to-phase dependency matrix. C = Critical, I = Important, Inf = Informational, D = Defining, G = Governing.

#	Title	F	S	P	H	SM
1	Architecture	C	C	I	I	I
2	Ta–Si Qubits	C	C	C	I	G
3	Error Corr.	I	C	C	C	G
4	Photonic Links	Inf	I	C	C	G
5	AI Decoding	Inf	I	I	C	C
6	Software	C	C	C	C	C
7	Cryo Infra.	Inf	I	I	C	C
8	Algorithms	Inf	Inf	I	C	C
9	Benchmarking	C	C	C	C	C
10	Roadmap	D	G	G	G	D

B. Key Cross-Paper Dependencies

1) Paper 2 gates Paper 3. QEC overhead is fundamentally determined by physical error rates. If D-3 and D-4 are not met, no QEC scheme achieves targeted overhead.
2) Paper 4 gates Paper 1 at L3+. Without transduction milestones I-1 and I-2, the architecture cannot scale beyond L2.
3) Paper 5 enables Paper 3 at scale. AI-accelerated decoding is required for decoder latency at Aggressive and Stretch scales.
4) Paper 7 constrains Paper 1. Cryogenic cooling capacity sets hard limits on qubit density per module.
5) Paper 9 validates all papers. Without reproducible benchmarks, no gate can be formally passed.
6) Paper 8 drives Paper 3 requirements. Algorithmic resource estimates determine the QEC overhead and code distance requirements.

VII. Integration and Decision Logic

A. Phase Transition Decision Tree

Each phase transition follows a structured decision process: (1) all gates met at Aggressive+ level leads to advancement on the Aggressive/Stretch track; (2) all gates met at Base level leads to advancement on the Base track with Stretch reassessment; (3) one or more gates failed activates fallback plans, with recovery time determining whether the phase is extended, the next phase scope is restructured, or the affected scenario is retired.

B. Scenario Transition Rules

- Stretch to Aggressive: Triggered if any subsystem cannot demonstrate a credible path to its Stretch milestone by end of PIONEER.
- Aggressive to Base: Triggered if QEC overhead remains above 500:1 at end of HORIZON, or no application benchmark shows meaningful speedup.
- Base is always maintained. The program delivers commercially relevant technology under all plausible outcomes.

C. Annual Review Cadence

TABLE XIX
Annual review schedule.

Review	Timing	Scope
Device	Q1	D-1–D-5; yield; fab status
QEC	Q2	Q-1–Q-4; decoder dev.; code selection
Systems	Q3	Integration; cryo; control scaling
Program	Q4	Scenario viability; budget; go/no-go

VIII. Comparison with Industry Roadmaps

A. IBM Quantum Roadmap

IBM's roadmap [1] projects 100,000-qubit systems through modular architectures. QONTOS shares the modular philosophy but differs: (a) QONTOS targets higher qubit counts via photonic inter-refrigerator networking; (b) QONTOS makes QEC overhead an explicit gate; (c) QONTOS defines explicit fallback architectures at each phase.

B. Google Quantum AI

Google's demonstration of error correction below the surface code threshold [2] validates the surface-code approach used in the QONTOS QEC subsystem. However, Google's demonstrated system (approximately 100 physical qubits for one logical qubit) corresponds to QONTOS Phase 2, not Phase 5.

C. Industry-Wide Context

Preskill's characterization of the NISQ era [10] remains relevant. The QONTOS roadmap is designed to transition from NISQ through early fault tolerance (Base) to large-scale fault tolerance (Stretch), recognizing that most of the industry will operate in the NISQ-to-early-FT regime through 2030.

[CLAIM-CONTEXT] McKinsey's Quantum Technology Monitor [9] projects that the quantum computing market could reach tens of billions of dollars by 2040. QONTOS is positioned to contribute at any scenario level.

IX. Workforce and Organizational Requirements

A. Team Scaling by Phase

TABLE XX
Team scaling by phase.

Phase	Size	Critical Capabilities
FOUNDATION	30–50	Architecture; simulation; benchmarking
SPUTNIK	80–120	Fabrication; packaging; QEC
PIONEER	150–200	Photonics; distributed systems; cryo
HORIZON	250–350	Applications; reliability; integration
SUMMIT	400–500	Operations; commercial; customer eng.

B. Critical Capability Gaps

1) Microwave-to-optical transduction expertise. Fewer than 200 researchers worldwide.
2) Cryogenic systems engineering at scale. Multi-refrigerator installations do not yet exist.
3) Fault-tolerant compilation. Must be developed in-house.

X. External Communication Framework

A. Communication Guardrails

TABLE XXI
Communication guardrails.

Type	Acceptable	Not Acceptable
Targets	"Our Stretch scenario targets…"	"We will build…"
Timeline	"Under aggressive assumptions, by 2030…"	"By 2030, we will deliver…"
Applications	"Our resource estimates indicate…"	"We will achieve advantage in…"
Comparison	"Our architecture shares principles with…"	"Our approach is superior to…"

[CLAIM-ROADMAP] All external communications must use scenario-qualified language. Unqualified claims about the Stretch scenario are not supported by this roadmap.

XI. Conclusion

This capstone paper defines the QONTOS program as a gated engineering plan rather than a single-path promise. The five-phase structure—FOUNDATION, SPUTNIK, PIONEER, HORIZON, and SUMMIT—provides a disciplined sequence of decision points, each governed by quantitative subsystem milestones with explicit fallback plans.

The Base scenario delivers a commercially relevant modular quantum computing platform under conservative assumptions. The Aggressive scenario requires major but plausible breakthroughs consistent with current research trajectories demonstrated by Google [2], IBM [1], and leading academic groups [3], [12]. The Stretch scenario of one million physical qubits and ten thousand logical qubits remains a conditional north star, achievable only if every subsystem meets its aggressive-case milestone on schedule.

Under aggressive but explicit assumptions regarding device coherence ($T_1 > 800$ µs, two-qubit gate error $< 10^{-3}$), QEC overhead reduction (effective ratio below 300:1), and interconnect fidelity (transduction efficiency above 15%, Bell pair fidelity above 97%), QONTOS could reach quantum advantage on selected applications within the 2030 timeframe. Full-scale advantage on flagship problems such as FeMoco simulation [5] or RSA-2048 factoring [7] remains a Stretch-scenario outcome contingent on the 100:1 overhead target and data-center-scale infrastructure.

[CLAIM-ROADMAP] The roadmap is feasible only if it remains gated, scenario-based, and tightly linked to benchmark evidence. The stretch scenario is not a commitment; it is a conditional engineering target that defines the program's maximum ambition while the Base and Aggressive scenarios ensure that every phase delivers tangible value.

The QONTOS program, as defined in this ten-paper series, represents a comprehensive and technically grounded approach to building large-scale modular quantum computers. Its credibility rests not on the boldness of its targets but on the rigor of its gates.

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